MODEL RESEARCH AND SELECTION

Healthcare Operations Intelligence Dashboard with Decision Analytics

A project document for data analytics, machine learning, and business intelligence

Database	MongoDB Atlas
Backend	Node.js and Express.js
Frontend	HTML, CSS and JavaScript
Visualization	Chart.js
Data Processing & Predictive Analytics	Python
Development Tools	MongoDB Compass and Visual Studio Code

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1. Introduction and Purpose

The Healthcare Operations Intelligence Dashboard is intended to bring hospital activity data—such as admissions, discharges, bed use, staffing, and treatment demand—into one interactive decision-support view. The dashboard combines historical reporting with analytical methods that can help teams understand current performance and prepare for operational changes.

This document focuses on model research and selection. It builds on the project's overall dashboard concept by reviewing suitable analytical techniques, comparing them against the project needs, and recording a recommended set of models for implementation.

The document has three practical aims:

- Clarify the operational questions the analytics should answer.
- Compare candidate methods using performance, interpretability, and implementation effort.
- Provide a traceable rationale for the selected model stack before development begins.

2. Project Requirements and Analytics Objectives

The solution is expected to provide three complementary levels of analytics rather than only display historical totals.

Analytics layer	Purpose
Descriptive	Summarize past activity, including admission and discharge patterns, workload, and bed utilization; compare similar departments or facilities.
Predictive	Estimate future demand and continuous measures such as occupancy or length of stay, and identify units that may face elevated operational risk.
Decision support	Highlight unusual activity and explain the factors behind model outputs so that managers can interpret results and consider appropriate action.

Functional requirements

- Forecast admissions and treatment demand over time.
- Estimate continuous operational indicators, including occupancy and length of stay.
- Assign hospitals or departments to Low, Medium, or High risk groups.
- Create segments that support operational comparison and benchmarking.
- Identify abnormal patterns, spikes, or delays that depart from usual activity.
- Explain predictions in terms that healthcare administrators can understand.

Implementation constraints

The proposed work is planned for a Google Colab/Python environment with moderate computing resources. The expected input is structured, tabular operational data rather than images or free-form text. Because the main audience is healthcare management, understandable outputs are treated as a key selection criterion alongside predictive performance.

3. Research Approach

Candidate methods were reviewed by task family: time-series methods for demand forecasting; regression and classification models for numerical predictions and risk labels; unsupervised learning for segmentation; anomaly-detection methods for unusual events; and explanation tools for model transparency.

The comparison considered five dimensions: suitability for tabular data, expected task performance, interpretability, implementation and tuning effort, and overlap with functions already provided by another candidate. The aim was to select a complementary set of techniques rather than one algorithm for every problem.

4. Candidate Models and Techniques Reviewed

4.1 Forecasting Future Demand

Methods considered: ARIMA, Prophet, exponential smoothing, and SARIMA.

Time-series techniques use historical admissions, discharges, and treatment-demand observations to estimate future values. They can help reveal trend and recurring patterns. SARIMA explicitly represents seasonal cycles, which may be useful if the dataset shows a stable weekly or seasonal pattern. However, it introduces additional configuration and should be treated as an extension when seasonality is supported by the data. A general time-series approach is therefore the baseline choice.

4.2 Predicting Continuous Operational Metrics

Methods considered: Random Forest Regression, XGBoost Regressor, and Linear Regression.

Measures such as bed occupancy, length of stay, and resource use can depend on interacting factors and may not have simple linear relationships. Linear Regression remains useful as a straightforward benchmark, while Random Forest Regression can model non-linear relationships and provide feature-importance summaries. XGBoost can perform strongly on structured datasets, but generally requires more tuning and is less straightforward to explain. Random Forest offers a practical balance for this project.

4.3 Classifying Operational Risk

Methods considered: Random Forest Classification, XGBoost Classifier, and Logistic Regression.

Administrators need a clear risk category as well as numerical indicators. Logistic Regression is a useful baseline but can be limited when relationships between operational factors and risk are complex. Random Forest Classification accommodates non-linear patterns and can be paired with feature-level explanations. XGBoost may offer performance gains in some settings, but its extra tuning needs and more difficult interpretation reduce its advantage for a management-facing tool. The final decision should be validated with suitable classification metrics and a clearly defined risk-labeling approach.

4. Candidate Models and Techniques Reviewed (continued)

4.4 Segmenting Hospitals and Departments

Methods considered: K-Means Clustering and Hierarchical Clustering.

Clustering can reveal groups of facilities or departments with similar operational profiles without requiring pre-labelled outcomes. K-Means is relatively efficient and creates a defined number of segments that can be communicated in a dashboard. Hierarchical Clustering can show nested relationships, but its dendrogram-based output can be harder to use in a routine operational view and may require more computation.

4.5 Detecting Operational Anomalies

Methods considered: Isolation Forest, DBSCAN, and Local Outlier Factor (LOF).

Sudden demand surges, delays, and unusual resource patterns may not be obvious from aggregate reporting. Isolation Forest is designed to isolate observations that differ from the majority and is a suitable candidate for a moderate-sized tabular dataset. DBSCAN is primarily a density-based clustering method, while LOF also targets unusual observations. Given the planned use of K-Means for segmentation, Isolation Forest provides a distinct anomaly-detection role with less functional overlap.

4.6 Explaining Model Predictions

Methods considered: SHAP and LIME.

An explanation layer is needed so users can understand which input factors contributed to a prediction. SHAP attributes contributions to individual features and can be applied to tree-based models such as Random Forest. LIME can produce local explanations too, but including both tools would add complexity without a clearly separate requirement. SHAP is selected as the primary explanation approach, subject to checking that its outputs are presented accurately and clearly.

5. Comparative Evaluation Summary

Requirement	Candidates	Main trade-off	Decision
Demand forecasting	ARIMA / Prophet / SARIMA	Seasonal models may help when seasonality is evident, but require extra tuning.	Time-series forecasting; SARIMA optional
Continuous metrics	Random Forest / XGBoost / Linear Regression	Boosting may improve performance; linear models are simpler but less flexible.	Random Forest Regression
Risk classification	Random Forest / XGBoost / Logistic Regression	Potential accuracy gains must be balanced against explainability and tuning.	Random Forest Classification
Segmentation	K-Means / Hierarchical	Hierarchical analysis is richer but more involved to communicate.	K-Means
Anomaly detection	Isolation Forest / DBSCAN / LOF	Isolation Forest has a distinct role and limits overlap with clustering.	Isolation Forest
Explainability	SHAP / LIME	One consistent explanation framework avoids duplication.	SHAP

6. Recommended Model Stack and Rationale

The recommended stack contains six complementary methods. Together, they address forecasting, numerical prediction, risk classification, segmentation, anomaly detection, and explanation without expecting one model to solve every task.

#	Selected technique	Expected project output	Analytics layer
1	Time-Series Forecasting	Estimated future admissions and treatment demand	Predictive
2	Random Forest Regression	Predicted occupancy, length of stay, and related continuous indicators	Predictive
3	Random Forest Classification	Operational risk category: Low / Medium / High	Predictive
4	K-Means Clustering	Groups of facilities or departments for benchmarking	Descriptive
5	Isolation Forest	Flags for unusual operational observations	Decision support
6	SHAP Explainability	Feature contributions that help explain predictions	Decision support

6.1 Why a combination is preferable

The project contains different analytical problem types. Forecasting, regression, classification, clustering, and anomaly detection have different goals, while explainability supports several of the other stages. A single algorithm cannot fulfil all these roles appropriately. Selecting one technique for each requirement gives the dashboard a clearer and more maintainable design.

Random Forest is used for both regression and classification because it can capture non-linear relationships while offering practical ways to inspect feature importance. SHAP supplies a common explanation approach for those predictive models. K-Means and Isolation Forest are included for separate unsupervised tasks, and all methods are suitable candidates for prototyping in the planned Python environment.

Implementation note: these are proposed selections, not a claim that the models have already achieved a particular accuracy. Model performance should be tested using the actual dataset, suitable train/test procedures, and task-specific evaluation metrics.

7. Alternatives Considered but Not Selected

7.1 Optional methods for later phases

- **XGBoost Regressor / Classifier:** possible alternatives if evaluation shows a meaningful performance improvement that justifies extra tuning and explanation effort.
- **Logistic Regression:** a simple, interpretable benchmark for comparison with the selected risk classifier.
- **Hierarchical Clustering:** useful if future work needs nested relationships between departments or hospitals.
- **Principal Component Analysis (PCA):** may help if many correlated variables make clustering or modelling difficult.
- **SARIMA:** a seasonal forecasting refinement if the dataset demonstrates reliable recurring cycles.

7.2 Methods not included in the core stack

- **LIME:** excluded to avoid maintaining two explanation methods where SHAP is the chosen framework.
- **DBSCAN:** not selected because the main segmentation role is assigned to K-Means and anomaly detection to Isolation Forest.
- **Local Outlier Factor:** overlaps with the planned anomaly-detection function.
- **Multiple similar ensemble models:** parallel variants should only be added if measured results justify their additional cost and complexity.

8. Proposed Analytics Pipeline

The proposed flow moves from operational records through preparation and modelling to an actionable dashboard.

Stage	Activity / output
1	Collect hospital operations data: admissions, discharges, beds, staffing, and treatments
2	Clean and preprocess the data
3	Explore distributions, trends, missing values, and relationships
4	Engineer useful model features
5	Forecast future demand with time-series methods
6	Estimate continuous metrics with Random Forest Regression
7	Classify operational risk with Random Forest Classification
8	Create operational segments with K-Means
9	Flag anomalies with Isolation Forest
10	Explain relevant predictions with SHAP
11	Translate findings into decision-support recommendations
12	Present key outputs through the dashboard

8.1 Mapping the Pipeline to Analytics Layers

Layer	Methods and dashboard role
Descriptive — What happened?	Exploratory analysis and K-Means summarize historical patterns and group similar operational units.
Predictive — What may happen?	Time-series forecasting estimates demand; Random Forest models predict continuous measures and risk categories.
Decision support — What needs attention?	Isolation Forest identifies unusual observations, while SHAP helps explain the drivers behind model outputs. The dashboard can present these findings for review.

9. Conclusion

This review maps the project's six core needs—demand forecasting, operational metric prediction, risk classification, segmentation, anomaly detection, and explainability—to methods suited to those tasks. The selection prioritizes a workable balance of analytical capability, transparency, and implementation effort for a healthcare management audience.

The recommended set is Time-Series Forecasting, Random Forest Regression, Random Forest Classification, K-Means Clustering, Isolation Forest, and SHAP Explainability. It supports the descriptive, predictive, and decision-support layers while keeping overlapping functions to a minimum. The recommendations should be confirmed through testing on the project dataset before the models are used for operational decisions.

10. References

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